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Acousto-Optic Modulation in Silicon Waveguides Based on Piezoelectric Aluminum Scandium Nitride Film

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Aluminum scandium nitride (AlScN) has attracted extensive attention for its excellent piezoelectric properties in the micro-electromechanical system applications. In this work, AlScN is demonstrated to be a promising candidate for on-chip acousto-optic coupling interactions with outstanding piezoelectric properties as well. Based on piezoelectric $Al_{0.6}Sc_{0.4}N$ film deposited on silicon-on-insulator platform, the proposed devices exhibit impressive acousto-optic coupling performances over a short interaction length of 210 µm with surface acoustic waves actuated by interdigital transducers. Meanwhile, the acousto-optic coupling performances are further improved in a compact spiral waveguide with eight interaction segments, where the measured modulation efficiency is enhanced from −21.5 to −12.5 dB and the radiofrequency (RF) 3-dB bandwidth is narrowed from 4 to 3.1 MHz at central frequency of 3.044 GHz with RF drive power of 18.6 dBm. The measured modulation efficiency is proportional to the incident RF drive power, depicting a linear increasing trend with a slope of one as the RF power increases up to 24.6 dBm. Such acousto-optic devices are capable of achieving broadband acousto-optic modulation and narrow-band microwave photonic filtering. Besides, the designed device structure provides a prospect in single-sideband modulation, acousto-optic frequency shift, and nonreciprocal light propagation.

1. Introduction

Aluminum scandium nitride (AlScN), scandium doped aluminum nitride (AlN), is an emerging piezoelectric material with excellent piezoelectric properties.[1–5] Due to the brilliant piezoelectric coefficients superior to AlN (d_{33} = 5 pC N^{-1}), AlScN (d_{33} = 25 pC N^{-1}) has attracted extensive attentions in micro-electromechanical system (MEMS) applications related with bulk acoustic waves (BAWs) and surface acoustic waves (SAWs),[6–9] such as acoustic sensors, energy harvesters and radiofrequency (RF) filters.[10–15]

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In recent years, piezoelectric materials, such as AlN, lithium niobate (LN) and GaAs, are frequently applied to acousto-optic (AO) coupling interactions with great potentials.[16–24] With a travelling SAW actuated by the interdigital transducer (IDT), the optical waves in waveguides are scattered by the travelling refractive index gratings and frequency-shifted through Doppler effect.[25] The AO coupling in piezoelectric materials offers a promising way to implement on-chip AO modulation, microwave photonic filtering, acousto-optic frequency shift, and nonreciprocal light propagation.[16,17,26,27] Different from the intrinsic optomechanical interactions in waveguides,[28–34] it is unnecessary to design suspended waveguides or optomechanical cavities for the AO interactions, which guarantees the stability and feasibility for on-chip applications.[16,35] Meanwhile, the electrically driven AO interactions exhibit higher AO scattering efficiency than the intrinsic optomechanical interactions, which can be adjusted in both optical domain and electrical domain.[36,37] Although LN (d_{33} = 6 pC N^{-1}) is a good candidate in AO interactions for its outstanding optical and piezoelectric properties,[38] it is not compatible with complementary metal-oxide-semiconductor (CMOS) technology to achieve large-scale fabrication.[20–22] As for AlN, it is generally deposited on silicon substrate through magnetic sputtering that is compatible with CMOS technology, but it poses a challenge to design waveguide structures in AlN layer because of the smaller refractive index than silicon substrate.[17,18] Therefore, suspended waveguide structures are indispensable for AlN on silicon substrate to carry out AO interactions. Surprisingly, a nonsuspended structure of AlN deposited on silicon-on-insulator (SOI) platform with a SiO_2 interlayer had realized AO modulation recently, where the optical wave is mostly confined in the nonpiezoelectric silicon layer instead of AlN layer.[39] In that structure, the SAW actuated by the IDT propagates from AlN layer across SiO_2 interlayer and finally modulates the optical waves in the silicon waveguides, where the SiO_2 interlayer greatly increases the SAW propagation loss and degrades the acoustic performances.[40] At last, with the same CMOS-compatible deposition as AlN and better piezoelectric properties than AlN,

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AlScN is an ideal candidate to implement AO coupling interactions on SOI platform.

In this article, AO modulation in silicon waveguides based on piezoelectric $Al_{0.6}Sc_{0.4}N$ film deposited on SOI platform is demonstrated. The piezoelectric $Al_{0.6}Sc_{0.4}N$ film is deposited upon the silicon waveguides so that the electrically driven SAWs through IDTs can directly modulate the optical wave in silicon waveguides without any extra attenuation. Furthermore, a compact structure of spiral waveguide is designed to improve the AO coupling performances. Then, a heterodyne detection experimental setup is used to measure and characterize the AO modulation of the proposed devices. Consequently, the fabricated devices represent broadband AO modulation and narrowband microwave photonic filtering.

2. Principle and Design

AO coupling results from elasto-optic effect.[41] When a strain field is applied on the waveguide, the material dielectric tensor is perturbed, leading to the change of the effective index of the optical modes in waveguide. As a result, the optical field in the waveguide can be modulated by the applied time-harmonic strain field (for details see Supporting Information section). The AO coupling factor Γ is defined as the spatial overlap of the SAW strain field and the involved optical modes over the cross-section of the waveguide,[16,42] given by

$$\Gamma = \frac{\iint E_{1x} E_{2x}^{*} n_0^2 \left(p_{11} S_{xx} + p_{12} S_{yy}\right) \mathrm{d}x\mathrm{d}y}{\left(\iint |E_{1x}|^2 \,\mathrm{d}x\mathrm{d}y \iint |E_{2x}|^2 \,\mathrm{d}x\mathrm{d}y\right)^{1/2}} \tag{1}$$

Here, n_0 is the equivalent refractive index of waveguide material without strain, p_{11} and p_{12} are the photoelastic coefficients, S_{xx} and S_{yy} are the xx-component and yy-component of the strain field. E_{1x} and E_{2x} are the x-component electric fields of the probe light and the scattered light, respectively.

The 3D model of designed AO device is shown in **Figure 1**a. After waveguide structures fabricated on the device layer of SOI platform, a piezoelectric $Al_{0.6}Sc_{0.4}N$ layer is deposited on it, followed by an IDT patterned upon the piezoelectric $Al_{0.6}Sc_{0.4}N$ layer. As shown in Figure 1b, a probe light with frequency ω_p can be scattered to the sidebands at frequency $\omega_p + \Omega$ and $\omega_p - \Omega$ by an incident SAW with frequency Ω and axial wavevector q, which requires phase-matching during the AO scattering process. According to the optical dispersion curve of the optical mode TE_0 in 750-nm wide and 70-nm depth etched ridge silicon waveguide, the difference of axial wavevector between the probe light and the optical sidebands is vanishingly small. In order to satisfy the phase-matching condition, the SAW axial wavevector along the optical propagation axis is also required to be vanishingly small. Therefore, the interdigital electrodes of the IDTs are designed parallelly to the optical propagation axis ($q = 0$) for convenience so that the AO coupling interactions are automatically in the quasi phase-matching state. And the wavevector mismatch Δq_{pm} can be represented as

$$\Delta q_{pm} = \beta\left(\omega_p + \Omega\right) - \beta\left(\omega_p\right) - q \approx \frac{\partial \beta}{\partial \omega}\Omega - q = \frac{\Omega}{v_g} \tag{2}$$

where v_g is the group velocity of the optical mode in waveguides at frequency ω_p. In this work, the group velocity of the optical mode is $v_g \approx 8.1 \times 10^7$ m s^{-1} at the optical wavelength

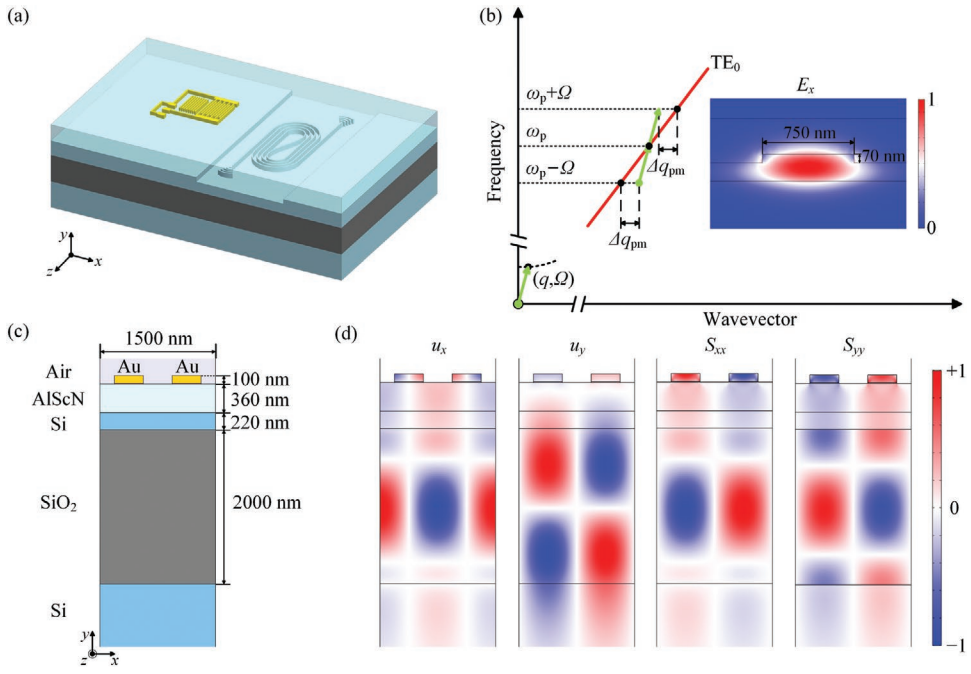


Figure 1. Design of AO devices on SOI platform. a) 3D model of the proposed AO device. b) Phase-matching diagram with dispersion curve of optical wave (red line) and incident acoustic wave (green arrow), together with the x-component electric field profiles of the optical mode TE_0 in the waveguide. Note that the scale of the diagram is not real. c) Simulation model of SAW actuated by the IDT on the SOI platform with periodic boundary conditions on the left and right boundaries. d) Simulation results of SAW displacement profiles and strain profiles in one period.

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Table 1. The electroacoustic parameters of $Al_{0.6}Sc_{0.4}N$ film used in the simulation, extracted from the reported experimental data.[7,9,43]

ρ [g cm^{-3}]	c_{11} [GPa]	c_{12} [GPa]	c_{13} [GPa]	c_{33} [GPa]	c_{44} [GPa]	e_{15} [C m^{-2}]	e_{31} [C m^{-2}]	e_{33} [C m^{-2}]	$\varepsilon_{33}/\varepsilon_0$
3.35	298	143	125	197	103	-0.20	-0.82	2.50	20.0

around 1550 nm, and the operating acoustic frequency Ω is about 3 GHz. Thus, the total value of phase mismatch with an interaction length of $L = 210$ µm is $\Delta q_{pm} L \approx 7.8 \times 10^{-3}$, so the AO coupling process is approximately phase-matched.

The designed finite element model shown in Figure 1c is to simulate the SAW profiles actuated by the IDT, and the electroacoustic parameters of $Al_{0.6}Sc_{0.4}N$ film used in the simulation are shown in **Table 1**. Beginning with 220-nm thick SOI platform, a layer of 360-nm thick piezoelectric $Al_{0.6}Sc_{0.4}N$ film is deposited upon it, and the interdigital electrodes made of Ni/Au with a period of 1500 nm and a thickness of 10/100 nm are patterned on the $Al_{0.6}Sc_{0.4}N$ film, where 10-nm Ni is used for enhancing adhesion. Importantly, it is prerequisite to apply periodic boundary conditions on the left and right boundaries so as to simulate the practical situation of hundreds of pairs of interdigital electrodes. Since the *x*-component electric field profiles of 750-nm wide ridge waveguide are symmetry, the SAW strain profiles are required to be anti-symmetry with a transverse period of 1500 nm to obtain the maximal AO coupling factor Γ, shown in Figure 1d.

3. Experiments and Results

3.1. Characterization of $Al_{0.6}Sc_{0.4}N$ Film

The crystal orientation of $Al_{0.6}Sc_{0.4}N$ film deposited on SOI platform is investigated by X-ray diffraction (XRD), with 2θ pattern and rocking curve shown in **Figure 2**a,b, respectively. The peak intensity of X-ray 2θ pattern appears at 36.6°, indicating that the $Al_{0.6}Sc_{0.4}N$ film is deposited along preferred crystal orientation (0002). Different from the AlScN film deposited on silicon substrate,[9] there is not any strong peak arising at 69° that reveals the crystal orientation (004) of silicon. In Figure 2b, the full-width-half-maximum (FWHM) of the $Al_{0.6}Sc_{0.4}N$ film in X-ray rocking curve is measured to be 2.9°, while the FWHM of the AlScN film deposited on silicon substrate is typically about 2°.[44] Therefore, the SOI platform is quite different from the silicon substrate for AlScN film deposition, mainly because the top layer of 220-nm thick silicon device is too thin. Thus, improving the crystal quality of the AlScN film deposited on SOI platform is an urgent issue to deal with, such as modifying the deposition condition, adding buffer layer between the silicon layer and the AlScN layer, as well as increasing the thickness of the AlScN film.[45]

The spectral ellipsometry test is also carried out on the deposited $Al_{0.6}Sc_{0.4}N$ film, with the measured refractive index *n* (black line) and extinction coefficient *k* (blue line) shown in Figure 2c. The measured extinction coefficient *k* is gradually approaching to zero as the optical wavelength increase from 700 to 1000 nm, so that we can use the Sellmeier formula to fit the refractive index *n* and extend the wavelength to 1700 nm (dash line), which is larger than the refractive index of AlN film due to the doping scandium. As a result, the refractive index *n* and extinction coefficient *k* of deposited $Al_{0.6}Sc_{0.4}N$ film are 2.274 and 0 at the optical wavelength of 1550 nm, indicating that the $Al_{0.6}Sc_{0.4}N$ film is a good candidate in optical applications.

3.2. Experimental Setup

A heterodyne detection method is adopted to distinguish and measure the modulated optical power of two sidebands, and the schematic diagram of the experimental setup is shown in **Figure 3**a. A pump light with frequency ω_p emitted from tunable laser (TL) is split into a probe light path and a reference light path through an optical coupler. The probe light is coupled into device under test (DUT) after controlling the polarization with a polarization controller (PC). With a RF signal at frequency Ω generated from RF signal generator (RFSG) loading

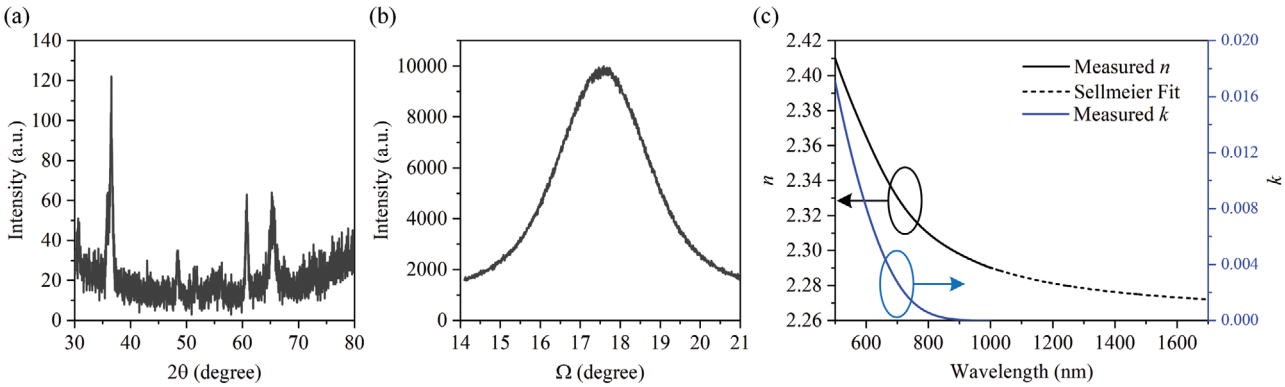


Figure 2. Characterization of $Al_{0.6}Sc_{0.4}N$ film deposited on SOI platform. a) XRD 2θ patterns. The peak intensity appears at 36.6° that shows crystal orientation (0002). b) XRD rocking curve with a FWHM of 2.9°. c) Measured refractive index *n* (black line) and extinction coefficient *k* (blue line) as a function of optical wavelength, together with a Sellmeier fitting curve (dash line).

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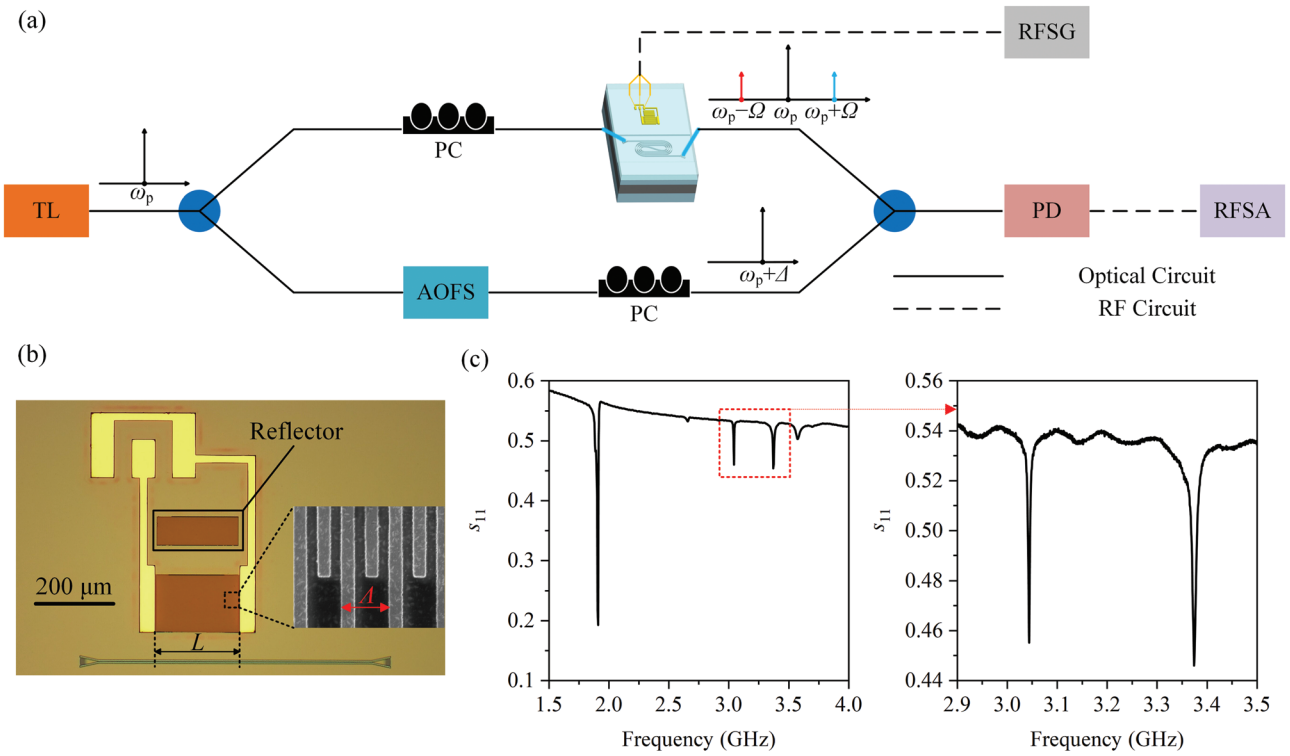


Figure 3. a) Schematic diagram of the heterodyne detection experimental setup. b) Microscope view of a typical fabricated AO device, consisting of an IDT with a SAW reflector and a straight silicon waveguide. Inset is a scanning electron microscope (SEM) image of the interdigital electrodes. c) Wideband frequency response of electrical reflection coefficient s_{11} of the IDT. There are three strong resonant valleys at 1.905, 3.044, and 3.375 GHz, respectively. The zoomed-in figure of the dash box is shown on the right.

on the GSG (ground-signal-ground) contact pads of IDT, part of the probe power is scattered to the sidebands at frequency $\omega_p + \Omega$ and $\omega_p - \Omega$. In the meantime, the reference light is frequency-shifted to $\omega_p + \Delta$ ($\Delta = 2\pi\times200$ MHz) using an acousto-optic frequency shifter (AOFS), and it is also controlled to the same polarization as the probe light through another PC. Then, the probe light path and reference light path are combined using another optical coupler and detected by a high-speed photodetector (PD). Finally, the resultant RF beat notes at frequency $\Omega - \Delta$ and $\Omega + \Delta$ output from PD are detected and displayed on a RF spectrum analyzer (RFSA), and the ratio of the modulated sidebands power and probe power at the output port of DUT can be directly measured by the RFSA, which is defined as the measured modulation efficiency $\eta_{mea} = P_{out}(\omega_p \pm \Omega)/P_{out}(\omega_p)$ throughout this article.

A typical fabricated AO device is shown in Figure 3b, including an IDT, a SAW reflector and a straight silicon waveguide. The IDT with GSG contact configuration contains 100 pairs of interdigital electrodes with periods of $\Lambda = 1500$ nm and an output aperture width of $L = 210$ μm, while another 50 pairs of interdigital electrodes with the same period serve as a SAW reflector patterned behind the IDT to reflect the backward SAW to forward. And a straight silicon waveguide stands in front of the IDT, perpendicular to the SAW propagation direction. Since the value of phase mismatch over the interaction length $\Delta q_{pm}L$ is extremely small, the intramodal scattering is quasi phase-matching over a large range of probe wavelength according to the optical dispersion curve. Figure 3c plots the wideband RF frequency response of electrical reflection coefficient s_{11} of the IDT, revealing that there are three strong SAW mode actuated at RF drive frequency $\Omega/2\pi$ of 1.905, 3.044, and 3.375 GHz. However, it is experimental confirmed that the actuated SAW mode at frequency of 1.905 GHz cannot obtain strong AO coupling strength with the probe wave. Hence, the SAW mode at frequency of 1.905 GHz is not taken into consideration in the next experiments, and the zoomed-in frequency response of electrical reflection coefficient s_{11} from 2.9 to 3.5 GHz is shown on the right of Figure 3c. According to the conversion efficiency formula from RF power to SAW power $\eta_{idt} = \Delta|s_{11}|^2$,[17] the conversion efficiency at RF drive frequency $\Omega/2\pi$ of 3.044 GHz and 3.375 GHz are 8.56% and 8.84%, respectively.

3.3. AO Modulation in a Straight Waveguide

The experimental results of AO modulation in a straight silicon waveguide are represented in **Figure 4**, with the schematic diagram shown in Figure 4a. The probe wave ω_p in the straight waveguide is scattered to the sidebands $\omega_p \pm \Omega$ over an AO interaction length of $L = 210$ μm.

The RF frequency response of the measured modulation efficiency is plotted in Figure 4b with a probe wavelength of $\lambda = 1550$ nm and a RF drive power of $P_{RF} = 20$ dBm. Note that there is 1.4-dB loss for 3-GHz RF signal along the 1-m transmission line from RFSG to the RF probe throughout this article, so that the actual RF power incident on the RF probe is 18.6 dBm.

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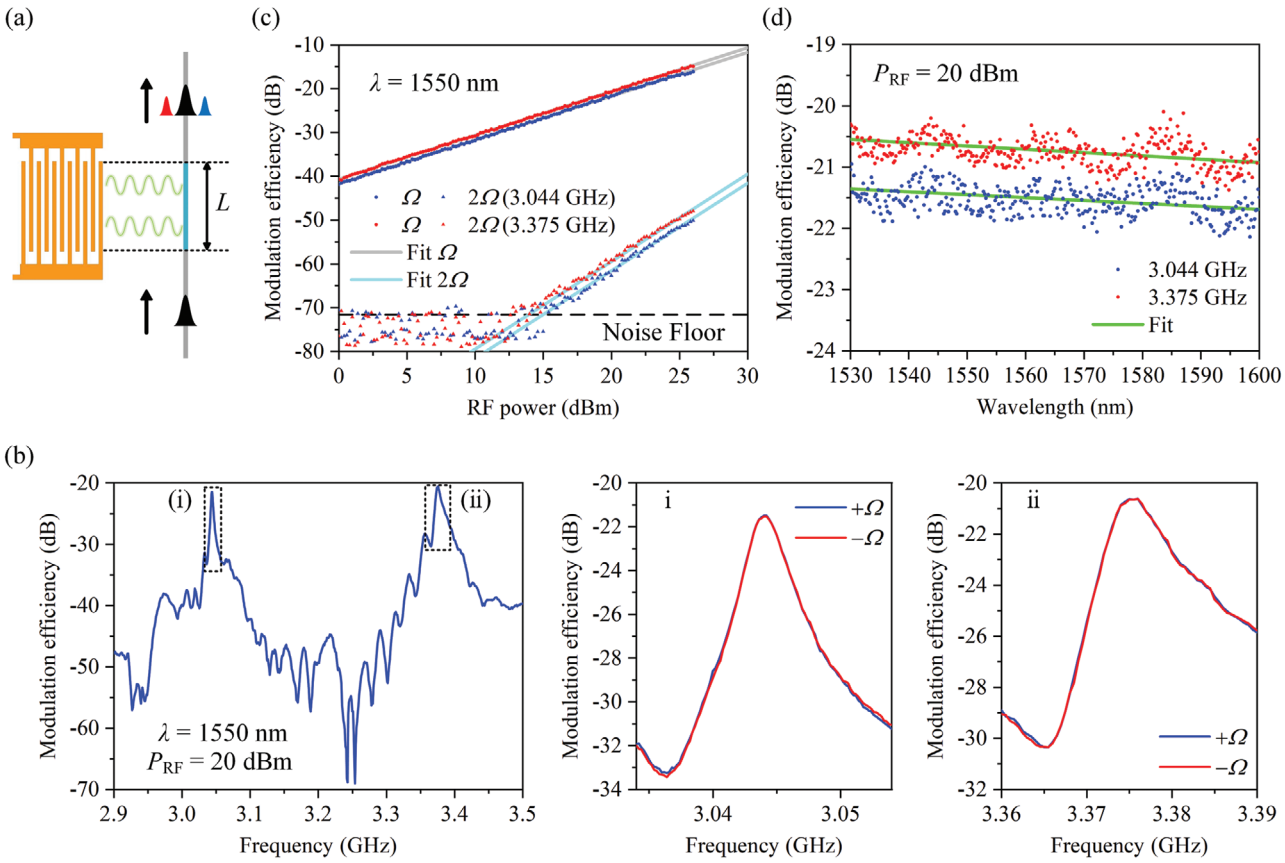


Figure 4. Results of AO modulation in a straight silicon waveguide. a) Schematic diagram of AO modulation over an interaction length of $L = 210$ µm. b) Relation between measured modulation efficiency and RF drive frequency. Inset (i) is the zoomed-in figure at central frequency of 3.044 GHz, and Inset (ii) is the zoomed-in figure at central frequency of 3.375 GHz. Note that there is 1.4-dB loss for 3-GHz signal along the 1-m transmission line from RFSG to RF probe. c) Relation between measured modulation efficiency and RF drive power with RF drive frequency inside parentheses. The modulation efficiency of Ω sidebands is linearly fitted with a slope of one (gray line), while the modulation efficiency of 2Ω sidebands is linearly fitted with a slope of two (cyan line). d) Relation between measured modulation efficiency and probe wavelength, together with linear fitting (green line) of the experimental data.

From Figure 4b, there are two strong peak of the measured modulation efficiency at the RF central frequency $\Omega/2\pi$ of 3.044 and 3.375 GHz, consistent with the resonant valleys of s_{11} in Figure 3c. Meanwhile, the zoomed-in figure at the central frequency $\Omega/2\pi$ of 3.044 GHz with a peak of −21.5 dB and a 3-dB bandwidth of 4 MHz is plotted in Inset (i), while the zoomed-in figure at the central frequency $\Omega/2\pi$ of 3.375 GHz with a peak of −20.6 dB and a 3-dB bandwidth of 11 MHz is plotted in Inset (ii), corresponding to figure of merit $V_\pi L = 1.06$ V cm for 3.044 GHz and $V_\pi L = 0.95$ V cm for 3.375 GHz, where V_π is the half-wave voltage. And it also suggests that the measured modulation efficiency of upper sideband +Ω is identical with the lower sideband −Ω, so it is sufficient to investigate intramodal AO scattering with one of the sidebands. Figure 4c shows the RF drive power dependence of the peak modulation efficiency at RF drive frequency $\Omega/2\pi$ of 3.044 and 3.375 GHz with a probe wavelength of $\lambda = 1550$ nm. At high RF drive power, cascade sidebands (2Ω) arise in the AO modulation process. However, due to the noise floor (≈ −100 dBm) of the RFSA, the modulated 2Ω sidebands are too weak to be detected at low RF drive power. As RF drive power increases up to 26 dBm, the increasing trend of Ω sidebands are linearly fitted with a slope of one (gray line), while the increasing trend of 2Ω sidebands are linearly fitted with a slope of two (cyan line), revealing a linear tendency for Ω sidebands and a quadratic tendency for 2Ω sidebands that fits well with the modulator theory.[39] Figure 4d depicts the relation between the measured modulation efficiency of Ω sidebands and the probe wavelength at RF drive frequency $\Omega/2\pi$ of 3.044 and 3.375 GHz with a RF drive power of $P_{RF} = 20$ dBm. Linear fitting (green line) is adopted to describe the variation trend of measured modulation efficiency with probe wavelength varying from 1530 to 1600 nm, where the data fluctuations (<1 dB) are mainly caused by the changing polarization of the reference light path during the process of wavelength tuning. Moreover, the linear fitting of measured modulation efficiency depicts a slightly decreasing trend in good agreement with AO scattering theory.[46]

3.4. AO Modulation in a Spiral Waveguide

In order to take full advantage of the SAW power and enhance the modulation efficiency, we also fabricate another AO device with a spiral waveguide presented in **Figure 5**a, and the

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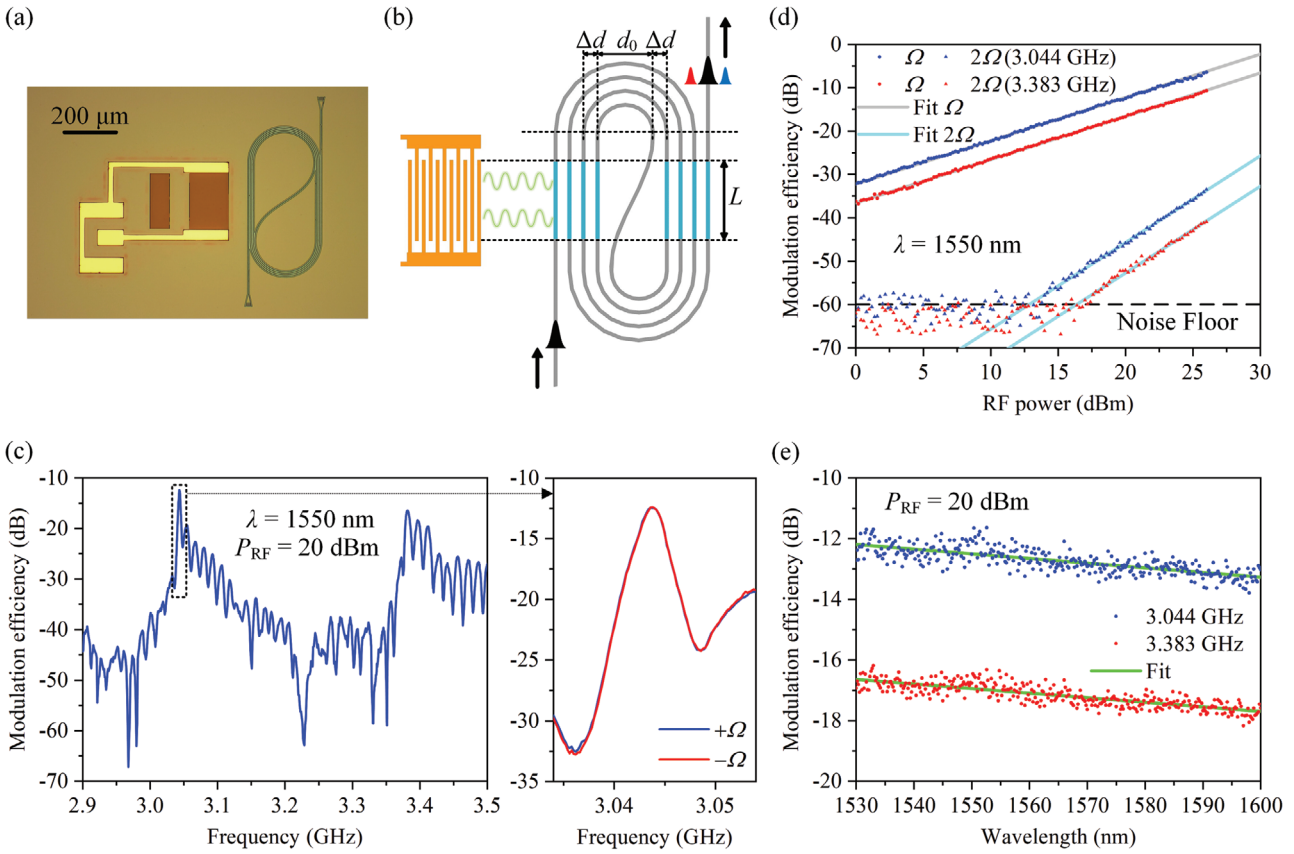


Figure 5. Results of AO modulation in a spiral waveguide. a) Microscope image of an AO device with a spiral waveguide. The area and total length of the spiral waveguide are 285 × 600 μm^2 and 6200 μm, respectively. b) Schematic diagram of the AO modulation in the spiral waveguide with eight interaction segments (blue stripes). c) Relation between measured modulation efficiency and RF drive frequency. The zoomed-in figure at central frequency of 3.044 GHz is shown on the right. d) Relation between measured modulation efficiency and RF drive power with RF drive frequency inside parentheses. The modulation efficiency of Ω sidebands is linearly fitted with a slope of one (gray line), while the modulation efficiency of 2Ω sidebands is linearly fitted with a slope of two (cyan line). e) Relation between measured modulation efficiency and probe wavelength, together with linear fitting (green line) of the experimental data.

schematic diagram is shown in Figure 5b. The area and total length of the compact spiral waveguide are 285 × 600 μm^2 and 6200 μm, respectively.

When the SAW propagates across the spiral waveguide, the probe wave in the spiral waveguide is scattered to the sidebands in the eight segments of interaction region (blue stripes), while the S-bend waveguide in the middle is not the active region of AO scattering due to the large phase mismatch. The optical amplitude of modulated sidebands in each interaction segment interferes with each other, and the resultant modulated optical amplitude at the output port is the superposition of modulated amplitudes in all interaction segments with phase difference. And the phase difference relative to the initial interaction segment contains both optical phase difference and acoustic phase difference (for details see the Supporting Information section III). Practically, due to the small wavevector difference of probe wave and modulated sideband wave, the impact of the optical phase difference (<0.073π) on the resultant optical amplitude of modulated sideband is so little in the spiral waveguide that it can be neglected. Besides, the minimum diameter d_0 and the transverse distance of adjacent interaction segment Δd are designed to be the integer times of the SAW transverse period Λ so as to control the acoustic phase difference of each interaction segment to be the integer times of 2π. Due to the large transverse distance (=184Λ) of the spiral waveguide compared with the SAW transverse period Λ, the resultant optical amplitude of modulated sideband is more sensitive to acoustic phase difference than optical phase difference.

The RF frequency response of measured modulation efficiency in the spiral waveguide, with a probe wavelength of λ = 1550 nm and a RF drive power of P_{RF} = 20 dBm, is plotted in Figure 5b together with a zoomed-in figure at central frequency $\Omega/2\pi$ of 3.044 GHz shown on the right. Due to the interference of the multi-segment interactions, there are many interference ripples in the frequency response of measured modulation efficiency. As a result, the strongest peak is –12.5 dB at RF drive frequency $\Omega/2\pi$ of 3.044 GHz with a 3-dB bandwidth of 3.1 MHz, while the second strongest peak is –16.7 dB at RF drive frequency $\Omega/2\pi$ of 3.383 GHz with a 3-dB bandwidth of 8 MHz. However, there is no strong peak appearing at RF frequency of 3.375 GHz, because it suffers from interference cancellation at this RF drive frequency. Consistent with the circumstances in the straight waveguide, with RF drive power increasing up to 26 dBm shown in Figure 5c, the growing

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Table 2. The AO coupling performances comparison of this work and other reported piezoelectric AO devices on silicon platform.

Ref.	Device structure	Acoustic frequency [GHz]	Acoustic conversion efficiency	SAW power loss coefficient [dB mm^{-1}]	Modulation efficiency [dB][a)]	RF 3-dB bandwidth [MHz]
[16]	AlN/SiO_2/Si	0.9	\[b)]	≈12	\	130
[17]	Suspended AlN/Si	4.82	3.9%	\	\	56
[39]	AlN/SiO_2/SOI	3.11	6.5%	21.1 ± 1.5	−25.2 dB	20
This work	$Al_{0.6}Sc_{0.4}N$/SOI	3.044	8.56%	13.2 ± 5.0	−21.5 dB	4
This work	$Al_{0.6}Sc_{0.4}N$/SOI	3.373	8.84%	13.2 ± 5.0	−20.6 dB	11

[a)]Modulation efficiency is calculated under the normalized experimental condition of probe wavelength λ = 1550 nm and actual RF drive power P_{RF} = 18.6 dBm over one interaction segment; [b)]"\" denotes no relevant data

tendencies of Ω sidebands and 2Ω sidebands in the spiral waveguide are linearly fitted with slopes of one (gray line) and two (cyan line), respectively. Besides, the measured modulation efficiency of modulated Ω sidebands is on a slight decrease trend with linear fitting (green line) as the probe wavelength increase from 1530 to 1600 nm shown in Figure 5d.

4. Discussion

We have demonstrated AO modulation with SAW based on piezoelectric $Al_{0.6}Sc_{0.4}N$ film in silicon straight waveguides and spiral waveguides on SOI platform with RF signals loading on the IDT. As shown in **Table 2**, we make an AO coupling performances comparison of this work and other reported piezoelectric AO devices on silicon platform. The piezoelectric AO devices in this work exhibit higher acoustic power conversion efficiency, lower SAW power loss coefficient, higher modulation efficiency, and narrower RF 3-dB bandwidth in one interaction segment. Note that the modulation efficiency is calculated under the normalized experimental condition of probe wavelength λ = 1550 nm and actual RF drive power P_{RF} = 18.6 dBm. Compared with the reported device structure of AlN/SiO_2/SOI with an interaction length of 240 μm,[39] the modulation efficiency in this work with an interaction length of 210 μm is promoted by 3.7 and 4.6 dB at frequency of 3.044 and 3.375 GHz, respectively. These enhancements are attributed to the better piezoelectric properties of $Al_{0.6}Sc_{0.4}N$ film and the absence of the SiO_2 interlayer that degrades the SAW performances in silicon waveguides.

Since the optical power of modulated sidebands is proportional to the square of interaction length L^2, the modulation efficiency can be improved by extending the interaction length and designing specific waveguides with multiple interaction segments. The fabricated compact spiral waveguide compared with the straight waveguide in this work greatly promotes the AO coupling performances with eight interaction segments. The modulation efficiency of the compact spiral waveguide is improved from −21.5 to −12.5 dB and the RF 3-dB bandwidth is narrowed from 4 to 3.1 MHz at frequency of 3.044 GHz under the same experimental condition. And the modulation efficiency of the spiral waveguide can be further enhanced significantly through strict phase control of each interaction segment. For a spiral waveguide with many interaction segments and long total length, the optical loss coefficient of the waveguide and the SAW power loss coefficient should be taken into consideration. The SAW propagation power loss coefficient in this work is calculated to be 13.2 ± 5.0 dB mm^{-1} for acoustic frequency at 3.044 and 3.375 GHz (for details see the Supporting Information section IV), smaller than the reported SAW devices on SOI platform.[35,39] And the total optical insertion loss for the spiral waveguide in this work is about 5.0 dB, indicating an optical loss coefficient of 8.06 dB cm^{-1} that much larger than the optical loss coefficient of silicon waveguides. The large optical loss coefficient is mainly caused by the low quality of $Al_{0.6}Sc_{0.4}N$ crystal growth on SOI platform with a FWHM of 2.9° in XRD rocking curve, which increases the optical scattering loss at the interface of silicon layer and $Al_{0.6}Sc_{0.4}N$ layer, especially in the bending regions. Therefore, promoting the deposition crystal quality of $Al_{0.6}Sc_{0.4}N$ layer on SOI platform to reduce the optical loss coefficient as well as the SAW propagation loss coefficient, is a quite effective way to enhance the performances of the AO devices.[7,9,45]

Another effective way to improve performances of AO devices is to modify the IDT through structural designs and impedance-matching optimizations.[1] For instance, impedance-matching optimizations increase the conversion efficiency from RF power to acoustic power, and the special design of unidirectional IDT contributes to couple more acoustic power to the optical waveguides. Besides, acoustic resonance structures, such as acoustic cavities, edge reflectors and suspended structures,[17,21] can be implemented to generate a strong strain field that couples to the optical field in the waveguides to strengthen AO coupling.

5. Conclusion

In summary, AO modulation in silicon waveguides based on piezoelectric $Al_{0.6}Sc_{0.4}N$ film deposited on SOI platform is demonstrated in this work. The AO devices exhibit impressive performances over a short interaction length, and are further improved by designing a compact spiral waveguide structure with multiple interaction segments. As a result, the weak correlation of probe wavelength is capable of implementing broadband AO modulation, while the narrow 3-dB bandwidth of RF frequency response is able to achieve microwave photonic filtering. Eventually, this work also provides an idea to realize single-sideband AO modulation, acousto-optic frequency shift, and nonreciprocal light propagation through intermodal AO scattering from the fundamental optical mode to a higher-order optical mode in the multimode waveguides.

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6. Experimental Section

Device Fabrication: Beginning with a 220-nm thick device layer on SOI wafer, the 750-nm wide silicon waveguides were patterned with electron beam lithography and etched with a depth of 70 nm through inductively coupled plasma etching. Then, a piezoelectric $Al_{0.6}Sc_{0.4}N$ layer with a thickness of 360 nm was deposited on the SOI wafer through magnetic sputtering. Finally, the IDTs with 100 pairs of interdigital electrodes and contact pads were patterned through electron beam lithography, followed by 10-nm Ni and 100-nm Au deposition with electron beam evaporation and lift-off process.

Measurements: The s_{11} electrical reflection spectrums of fabricated IDTs were measured by a vector network analyzer (Anritsu 37369D) via a RF probe (GGB Industries). Before measuring s_{11} of IDTs, a calibration substrate (GGB Industries CS-5) was used to calibrate the impedance of cables and RF probe. The modulation efficiency was measured through a heterodyne detection experimental setup shown in Section 3.2 of the main text. The coupling loss of grating couplers between optical fiber and DUT was 6.5 dB each facet at the probe wavelength of 1550 nm.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

acousto-optic devices, aluminum scandium nitride, piezoelectric films, silicon waveguides

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